CLASS 9 — MATHS · FORMULA SHEET

Chapter 9: **Propositions and their Converses** | Quick revision, one page

If-then statements

- A proposition 'If P, then Q' has hypothesis P (the part after 'if') and conclusion Q (the part after 'then').
- Statements like 'All squares are rectangles' can be rewritten as 'If a quadrilateral is a square, then it is a rectangle'.

Converse

- The converse of 'If P, then Q' is 'If Q, then P' (hypothesis and conclusion are interchanged).
- The converse of a true statement may be true or false and must be checked separately.
- Example: 'If a number is divisible by 10, its ones digit is 0' and its converse are both true; 'If a number is divisible by 6, it is divisible by 3' has a false converse.

Counterexamples

- A single example that satisfies the hypothesis but fails the conclusion disproves a statement.
- Checking examples never proves a general statement; a logical proof is needed.
- Example: $n^2 > n$ fails for $n = 1$; 'every prime is odd' fails for 2.

Simple proofs

- Even number: $2k$; odd number: $2k + 1$; multiple of 3: $3k$ (for integers k).
- Sum of two even numbers is even; product of two odd numbers is odd: $(2a + 1)(2b + 1) = 2(2ab + a + b) + 1$.
- Proof layout: state what is given, use known facts step by step, state the conclusion.

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